Tectonophysics, 201 (1992) 121–140
Elsevier Science Publishers B.V., Amsterdam

Palaeomagnetism and $^{40}Ar/^{39}Ar$ dating of the Trégor dolerites (Armorican Massif, France)

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(Received July 3, 1990; revised version accepted November 20, 1990)

ABSTRACT

Ruffet, G., Perroud, H. and Féraud, G., 1992. Palaeomagnetism and $^{40}Ar/^{39}Ar$ dating of the Trégor dolerites (Armorican Massif, France). In: H. Perroud and G. Bylund (Editors), Palaeomagnetism of Old European Rocks. *Tectonophysics*, 201 (spec. sect.): 121–140.

The Trégor doleritic dike swarm intrudes the Cadomian North Trégor Batholith (Armorican Massif, France). Alternating field (A.F.) and thermal demagnetization performed on 158 samples from 18 dolerite dikes and 37 samples from 4 baked contact sites reveal a multi-component magnetization and a complex magnetic mineralogy which was thoroughly investigated with thermomagnetic experiments and polished thin-section examinations. Three magnetization components were recognized, in addition to a low-temperature component with a viscous origin. The most frequently observed component, named B ($D = 214°$, $I = 10°$), was isolated at 12 sites, independent upon their magnetic mineralogy, and appears to be a remagnetization (negative baked contact test). A Carboniferous age of acquisition is inferred from comparison of the B-component palaeopole (29° S, 318° E) with an Armorican APWP. At some sites, after demagnetization of the B-component, higher-temperature and coercivity components were isolated. Their directions fall into two groups, named A and C. Positive baked contact tests were obtained for both. The C-component ($D = 174°$, $I = 50°$), whose magnetic carrier is magnetite, yields a palaeopole (11° S, 2° E) which is similar to some Ordovician palaeopoles from the British Isles, south of the Iapetus suture, or Eocambrian palaeopoles from the Trégor area. The A-component ($D = 191°$, $I = -33°$), whose magnetic carrier is magnetite, partly transformed to leucoxene, yields a palaeopole (58° S, 336° E) and a palaeolatitude (17.9° N) not previously observed for the Armorican Massif in Palaeozoic time. Comparison between this palaeolatitude and those for northern Spain from Ordovician to Carboniferous time suggests a late Silurian remanence age. This interpretation supports the model of apparent polar wander for Gondwana between Ordovician and Devonian time proposed by Van der Voo, and Bachtadse and Briden. $^{40}Ar/^{39}Ar$ dating of mineral separates from the dikes do not allow us to place constraints on the acquisition age of the A- and C-components.

Introduction

Northern Brittany is characterized by a Precambrian basement, structured during the Cadomian orogeny and moderately affected by the Hercynian orogeny. It is intruded by a large number of doleritic dikes, concentrated in three major swarms (St Malo, Fougères and Trégor). In the St Malo doleritic swarm, all dikes are oriented approximately N–S. They have been previously the object of a K–Ar and palaeomagnetic study (Perroud et al., 1986a), which yielded a well defined Lower Carboniferous (330 ± 10 Ma) palaeomagnetic pole for the Armorican Massif. A palaeomagnetic study of the Trégor doleritic swarm was of great interest as (1) the dikes, unlike the N–S orientations of the St Malo and Fougères swarms, are oriented approximately E–W, and (2) preliminary K–Ar measurements revealed two age clusters around 350 Ma and 440–450 Ma, which suggests that there might have been two or more phases of dike emplacement. Furthermore, palaeomagnetic data for the interval Upper Ordovician–Lower Carboniferous are not available for the Armorican Massif, whereas data from this age are essential to test the new APWP proposed by Van der Voo (1988) and Bachtadse and Briden (1990) for Gondwana, which depends heavily on

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the Silurian Air palaeopole (Nigeria) (Hargraves et al., 1987) that drastically modifies the mid-Palaeozoic palaeogeography.

Geological setting

The deformation of the Trégor region (Fig. 1) took place during Hercynian tectonogenesis with the development of two major faults trending E–W. The northern-most (Tréguier fault; Auvray, 1979) separates the southern and northern Trégor domains. The Trégor dolerites constitute a dike swarm of exceptional density, strictly confined to the northern domain, which cut all formations including the Cadomian North Trégor granitoid batholith. The Trégor doleritic dikes, unlike other North-Armorican dolerites (generally oriented N–S) have an E–W trend (80–110°).

A petrographical study of the North Brittany dolerites presented by Velde (1970) defined several types of dikes. The Trégor dolerites belong to the two main groups, the uralitized dolerites and the dolerites with quartz and micropegmatite. They are characterized by the paragenesis plagioclase + clinopyroxene + chlorite + opaque minerals and are associated to hot spot tholeiites in an anorogenic continental domain (Auvray, 1979).

The radiometric age of the Trégor dolerites is not well constrained. Preliminary results (unpublished) obtained with the K/Ar method seem to indicate two intrusion periods, at 440 Ma and 350 Ma. The existence of dikes of different ages is not supported by field observations.

The dolerites frequently present a mineralogical destabilization, characterized by the appearance of secondary amphibole and chlorite, sericite, epidote and calcite. These accessory minerals are probably associated with a hydrothermal phase assumed to be contemporaneous with the intrusion (Auvray, 1979).

Experimental procedures

Palaeomagnetism

Natural remanent magnetization measurements were performed with a digital Schonstedt DSM-1 spinner magnetometer. Thermal (TH) demagnetization was carried out in a non-magnetic Schonstedt TSD-1 furnace with cooling in a zero field chamber (residual field < 10 nT). Alternat-

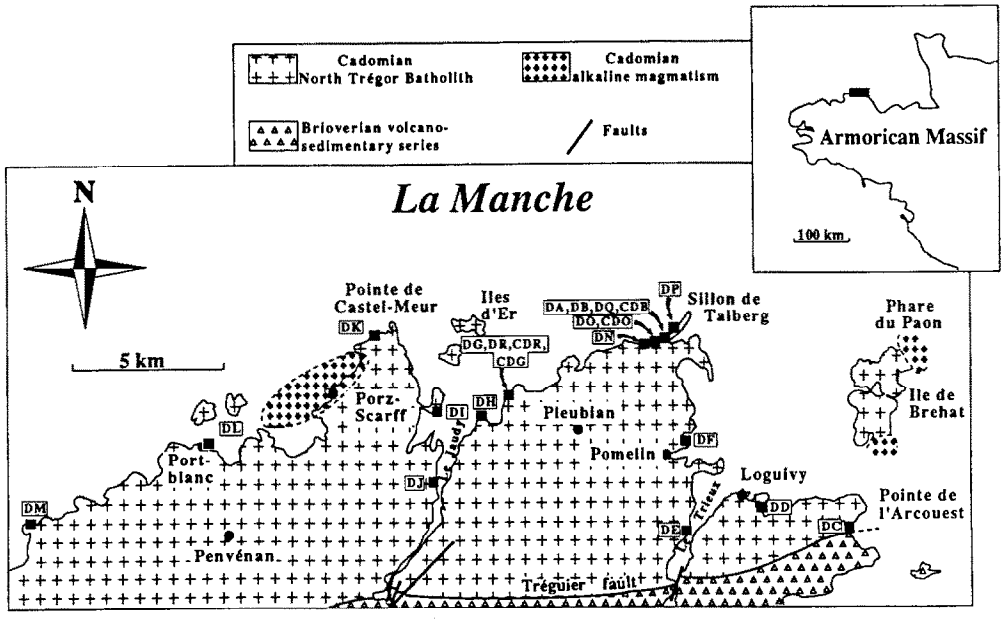


Fig. 1. Location of the study area (inset) and schematic geological map of the Trégor region. Sampling sites are shown by black squares.

ing field (A.F.) demagnetization was performed with a Schonstedt GSD-1. The residual field during A.F. demagnetization was also lower than 10 nT.

Geochronology

For stepwise heating experiments, the 80–125 μm fractions of plagioclase were separated by the standard methods (magnetic separators and heavy liquids). The final mineral separates were obtained by hand-picking. The samples were irradiated in the Osiris reactor (C.E.N. Saclay) with the monitor MMhb1 (520.4; Samson and Alexander, 1987) and with a total flux of 9×10^{18} n.cm^2 (rotated during irradiation). The standard $^{40}Ar/^{39}Ar$ step heating analyses (procedure described by Féraud et al., 1982) were performed with a radio frequency fusion system. The $^{40}Ar/^{39}Ar$ laser microprobe analysis procedure is described by Scaillet et al. (1991) and Ruffet et al. (1990). The mass spectrometer system includes a 120° M.A.S.S.E. tube, a Baür-signer source and an E.T.P. 1000 electron multiplier.

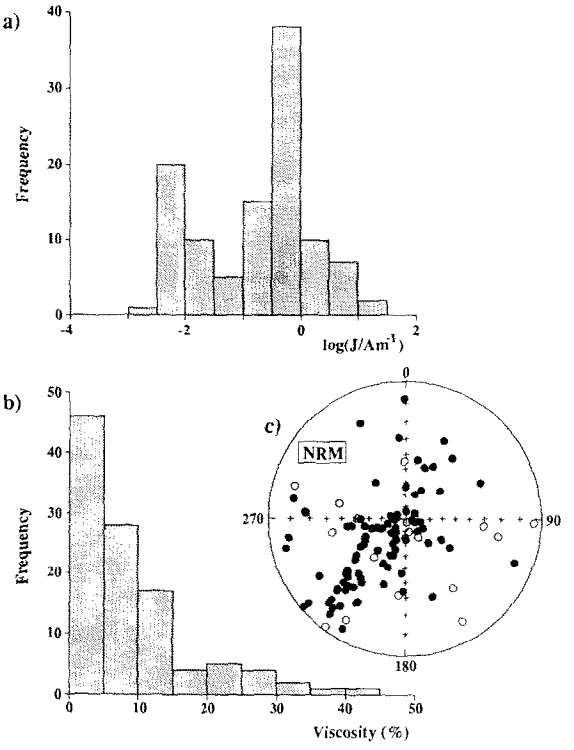


Fig. 2. (a) Frequency diagram of the NRM intensities; (b) frequency diagram of the viscosity coefficients; and (c) stereographic projection (in-situ coordinates) of the NRM directions.

Palaeomagnetic study

Sampling

162 samples from eighteen sites (Fig. 1) were drilled in the field and oriented with magnetic and sun compasses (weather permitting). Orientation errors are estimated to be about ±2° (Jones, 1978). Four sites were sampled in the host rock, the North Trégor batholith, for the purposes of contact tests. As the doleritic dikes have only suffered Hercynian fracturing (N–S), tilt corrections are not necessary.

Natural remanent magnetization

A short-period viscosity test (8 days; Thellier and Thellier, 1959) has been applied to 109 specimens taken from 96 cores. Specimens from the same core do not show significant differences in the results. The intensities of the natural remanent magnetization (NRM; Fig. 2a) show a bimodal distribution with maxima at around 5×10^{-3} A/m and 0.5 A/m. The viscosity coefficient (ratio of the viscous remanent magnetization (VRM) to the NRM; Fig. 2b) is rather low with a maximum value lower than 45%. Moreover, 84% of the specimens have viscosity coefficients lower than 15%. The highest values of viscosity generally correspond to samples with the lowest intensities (at sites DA, DB, DF and DQ). NRM directions are loosely scattered between the present-day field and a south–west direction (Fig. 2c).

The bulk susceptibilities also define two groups characterized by low (around 5×10^{-4} SI) or high (around 5×10^{-2} SI) values that seem directly correlated with the two groups of NRM intensities.

Anisotropy of magnetic susceptibility

The anisotropy of magnetic susceptibility (AMS) of 33 specimens was measured using the

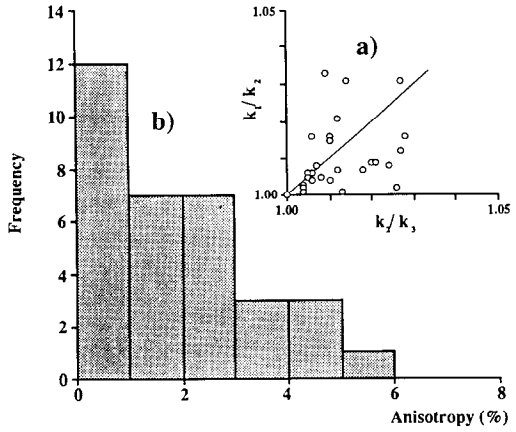


Fig. 3. Anisotropy of the magnetic susceptibility: (a) Flinn diagram (1965) for all measured specimens; (b) frequency plot of the anisotropy percentages.

Digico anisotropy delineator (Cogné, 1988). The total degree of anisotropy is lower than 5% for 97% of the measured specimens (Fig. 3b). Furthermore, the main susceptibility ratios (k_1/k_2 and k_2/k_3), plotted in the Flinn diagram (1965) (Fig. 3a) are quite variable. It appears that the Trégor dolerites do not present a significant AMS and are thus suitable for a classical palaeomagnetic study.

Demagnetizations

Thermal demagnetization (161 specimens) was applied to at least one specimen from each core. Alternating field demagnetization (87 specimens) was attempted on samples from each site, but was only applied systematically to samples from sites DA, DB, DC, DD, DE, DF, DG, DO, DP, DQ and DR because of the peculiar behaviour of specimens from other sites during the A.F. procedure.

After removal, below 300 °C or 20 mT, of a soft component attributed to viscous or present-day field parasitic magnetization, we have observed three groups of samples according to their demagnetization trajectories (illustrated in Figs. 4–6).

The first group (Fig. 4) is characterized by the presence of a high blocking-temperature or high-coercivity magnetization component, southwards and steeply inclined downwards and with sometimes an intermediate temperature or coercivity component, southwestwards with a shallow inclination.

The second group (Fig. 5), after removal of a component of intermediate blocking temperature or coercivity, directed shallowly downwards to the southwest, shows generally a high-temperature and coercivity component, directed southwestwards and intermediately inclined upwards.

The third group (Fig. 6) shows a high blocking temperature and coercivity component, southwestwards and shallowly inclined downwards, with frequently magnetization that does not decay towards the origin of the orthogonal demagnetization diagrams. This may mean that (1) there is parasitic magnetization (ARM) acquired during A.F. demagnetization or mineralogical changes provoked during thermal treatment, or (2) there is another magnetization which has not been resolved.

All the samples from site DM and some samples from site DP show high intensities, univectorial decay of the magnetization and characteristic directions inconsistent within the sites. This behaviour could be attributed to secondary isothermal remanent magnetizations (IRM) probably as a result of lightning.

Rock magnetism

The rock magnetism experiments performed to analyze the carriers of the magnetizations reveal a complex magnetic mineralogy:

(1) Thermomagnetic experiments (Fig. 7) using a Curie balance reveal two types of behaviour named respectively P and M.

The P-type, observed for sites DA, DB, DE, DF, DL, DM, DO, DP and DQ, shows principally, during the heating-phase of the experiment, paramagnetism (e.g. 86DB014A) with, nevertheless, frequently inflections around 300–350 °C and 550–580 °C (86DF051B and 86DL118). On the other hand, the cooling part of the experiment defines only one curie point around 540 °C. This behaviour characterizes samples with the lowest NRM intensities (2.8×10^{-3}–7.2×10^{-1} A/m).

The M-type, observed for sites DC, DD, DG, DH, DI, DJ, DK and DR, shows a well defined curie point at around 570–580 °C (e.g. 86DC25A). As for the P-type, inflections around 350 °C are frequently observed (86DH73B). This behaviour characterizes samples with the highest NRM intensities (1.5×10^{-1}–15.1 A/m).

(2) The isothermal remanent magnetization (IRM) (Fig. 8) was measured for one specimen from each P-type site. Saturation was usually

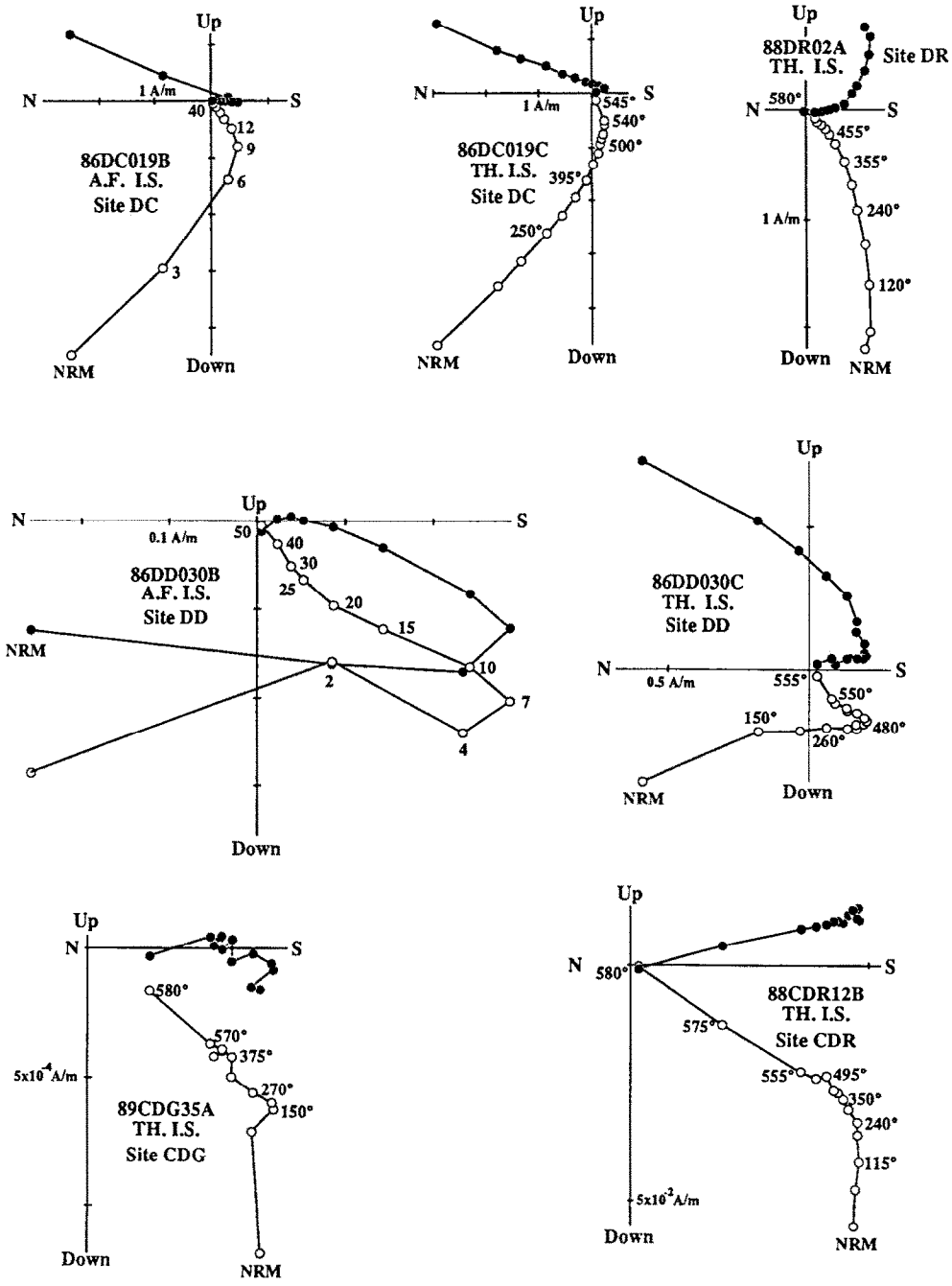


Fig. 4. Demagnetization diagrams (Zijderveld, 1967). Solid circles represent projections on the horizontal plane and open circles those on the vertical plane. See text for explanations. *TH.* = thermal demagnetization; *A.F.* = alternating field treatment; *I.S.* = in-situ coordinates.

obtained for applied direct fields between 150 and 300 mT (e.g. 86DA007B), but sometimes required 350–400 mT (e.g. 88DO03B).

These results suggest that (i) for the M-type sites, magnetite is the principal magnetic carrier, (ii) titano-magnetite seems to be the high-temperature carrier of magnetization at the P-type sites, and (iii) there is frequently, in all the sampled sites, a second magnetic carrier with a pseudo-"Curie temperature" around 350 °C. This secondary magnetic carrier is probably maghemite as suggested by the difference between the cooling curves of the two thermomagnetic experiments performed on sample 86DH073B, respectively under vacuum conditions (reducing environment) and at atmospheric pressure (oxidizing environment). Effectively, above 350 °C, maghemite converts to haematite that remains stable in an oxi-

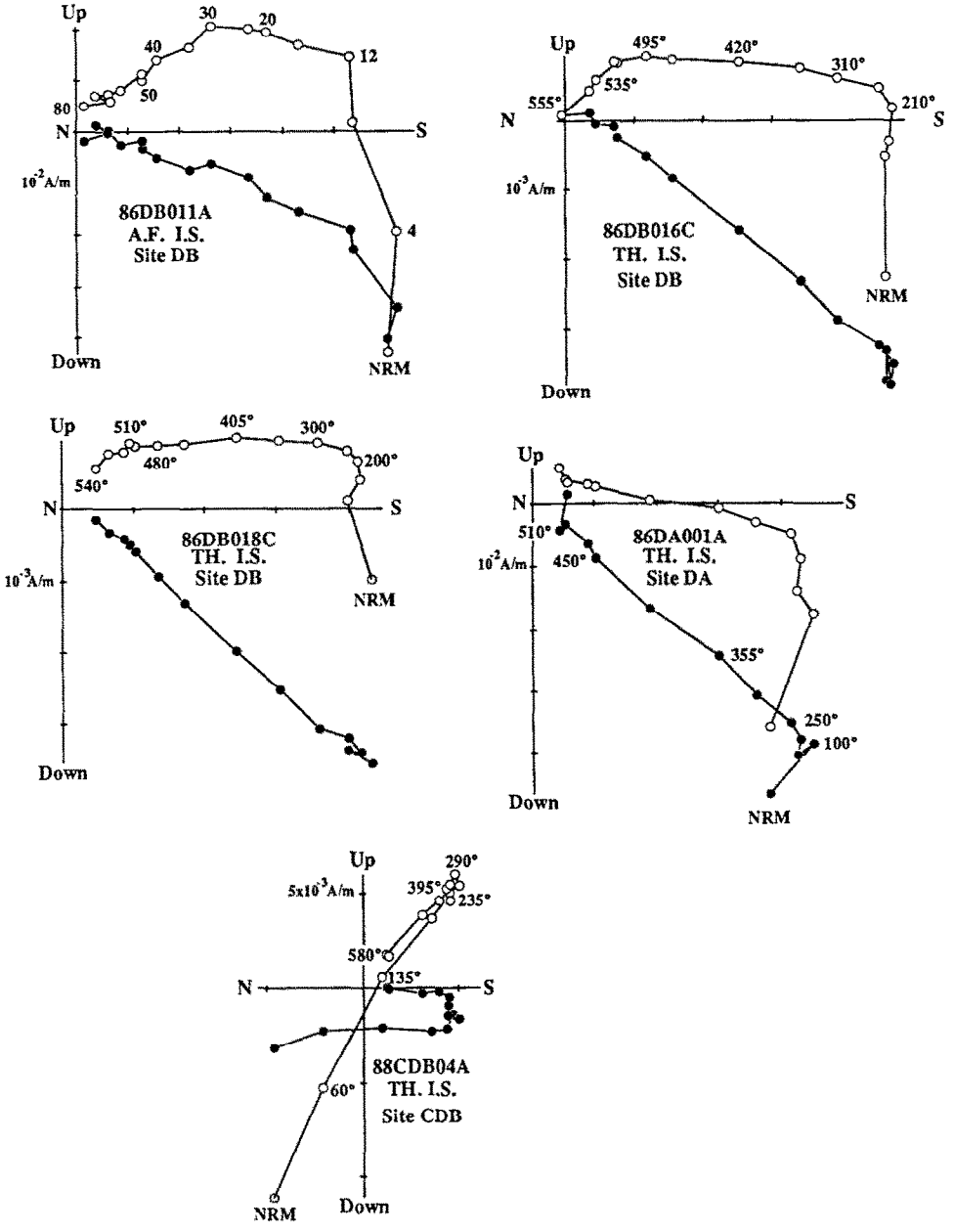


Fig. 5. Demagnetization diagrams (as in Fig. 4).

dizing environment or is transformed to magnetite under reducing conditions. As the magnetic susceptibility of the magnetite is higher than that of maghemite, which in turn is higher than that of haematite, the final susceptibility is higher or lower than the initial susceptibility, according to the environment conditions.

(3) The examinations in reflected light microscopy of thin sections for each site reveal also this duality.

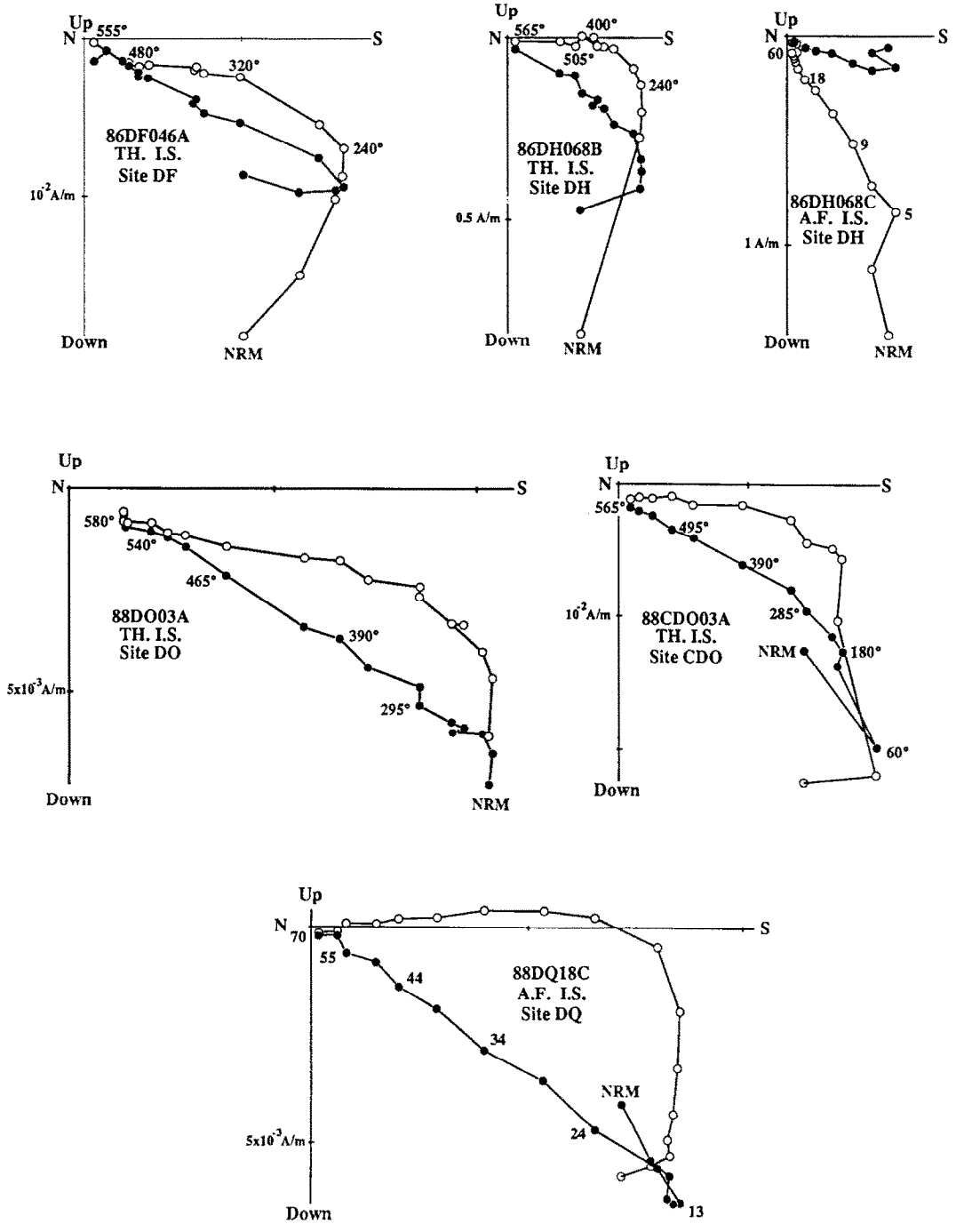


Fig. 6. Demagnetization diagrams (as in Fig. 4).

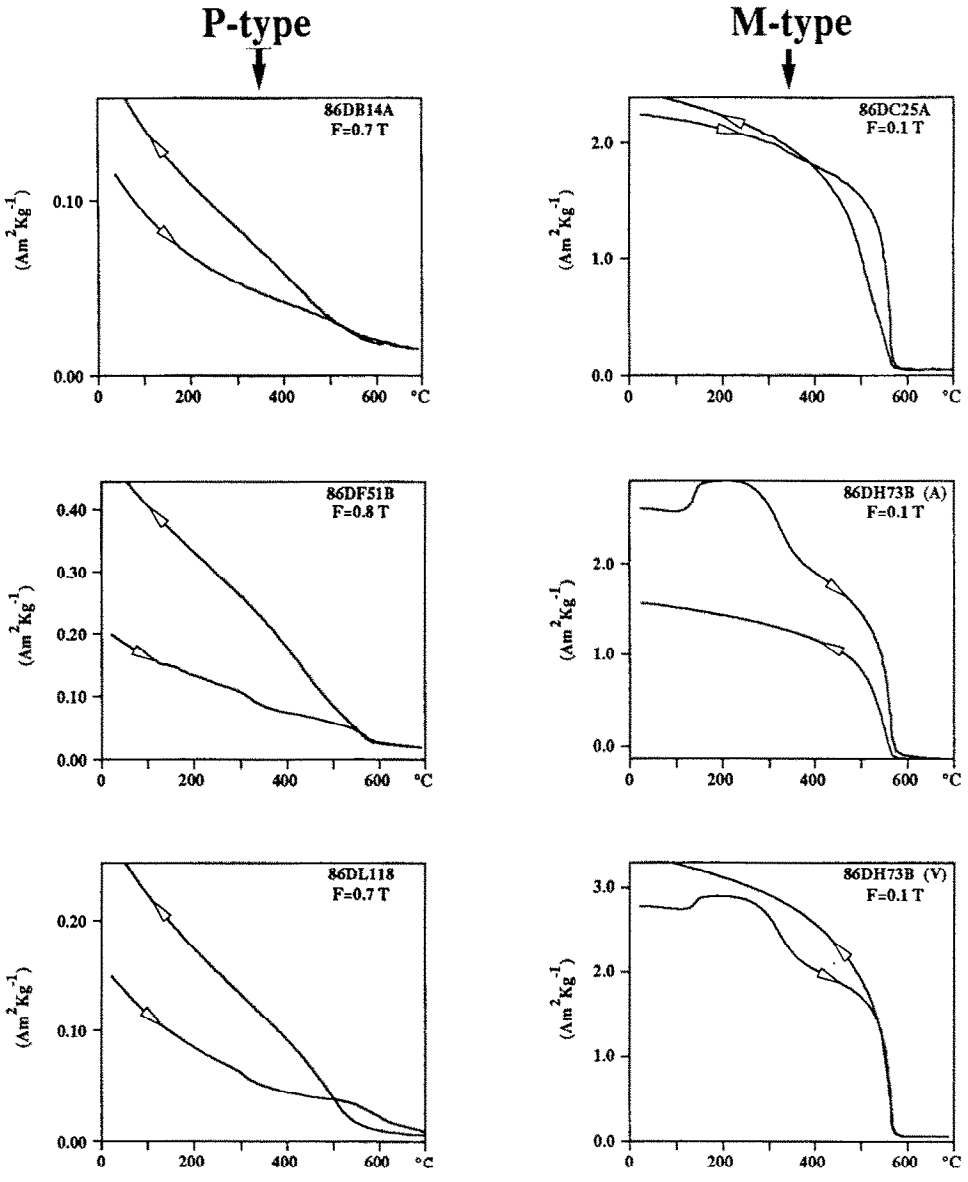


Fig. 7. Thermomagnetic analysis using a Curie balance.

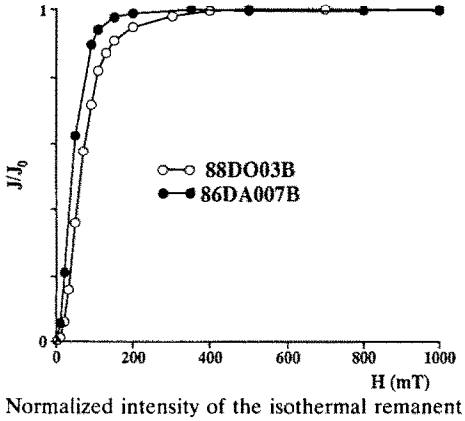


Fig. 8. Normalized intensity of the isothermal remanent magnetization (IRM) versus the applied field (in mT).

The P-type sites have a titano-magnetite mineralogy. The magnetic minerals are of large size and generally completely transformed to leucoxene. The destabilization results in relic titano-magnetite or ilmenite lamellae. Pyrite is always observed.

The M-type sites always show large crystals of magnetite, apparently without Ti except for site DC where lamellae of titano-magnetite or ilmenite are observed. Secondary magnetic minerals were occasionally observed (goethite, maghemite and pyrite).

The presence of maghemite can explain differences occasioned in demagnetization trajectories between two specimens of a same drilled-core that experienced A.F. and thermal demagnetizations (e.g. 86DH068B and 86DH068C; Fig. 6).

Magnetization components

The demagnetization trajectories were analysed from Zijderveld diagrams and the characteristic components of magnetization were determined graphically by line fitting.

Some characteristic demagnetization diagrams are presented in Figs. 4–6 whereas the results are compiled in Fig. 9 and summarized in Table 1.

The C-component (Fig. 4)

At site DC, a coherent high blocking temperature and coercivity magnetization is isolated, southward and steeply inclined. Thermal demagnetization, contrary to A.F. treatment (e.g. 86DCO19B), does not completely remove the low-coercivity magnetization component (e.g. 86DC019C). Nevertheless, the good convergence of great-circles determined from demagnetization trajectories (Fig. 9), allowed us to overcome this analytical problem. A similar magnetization component has been isolated in site DR (e.g. 88DRO2A) and in site DD, where it is poorly constrained by thermal treatment (e.g. 86DD030 C), but well isolated by A.F. demagnetization (e.g. 86DD030B). Some samples from site DF (e.g. 86DF046A) show a high-temperature component ($> 500\,^{\circ}$C) ($D = 213$, $I = 27$) that probably results from an imperfect separation between the C-component and a lower temperature or coercivity magnetization component (B-component, see below).

Host rocks of dike-sites DR and DG were sampled for baked contact tests (sites CDR and CDG). The characteristic magnetization component of the unbaked host rocks is directed intermediately downwards to the ESE (Ruffet et al., in prep.). In contrast, near the dike contacts (e.g. 89CDG35A and 88CDR12B), the characteristic magnetization direction of the host rock is similar to that of the dike-site DR, with unblocking temperatures up to $580\,^{\circ}$C. So, the baked contact test is positive.

The mean direction, computed from the three

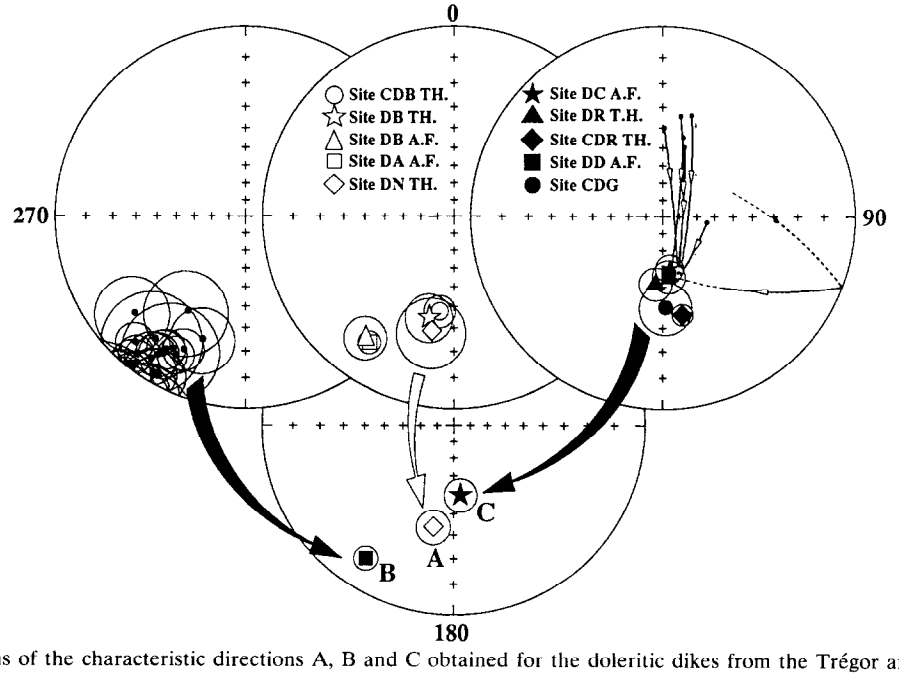


Fig. 9. Site-means of the characteristic directions A, B and C obtained for the doleritic dikes from the Trégor area and from the baked-contact sites (stereographic projection, in-situ coordinates). For the C-component are plotted the parts of the great circles determined from the demagnetization trajectories for samples from site DC. The solid lines represent the effective data path.

doleritic dike-site means (DC, DD and DR) and from the 2 baked contact sites means (CDR and CDG) is $D = 174.4$, $I = 49.5$, $k = 58.8$, $\alpha_{95} = 8.8$. This magnetization component will be labelled C-component. The corresponding magnetic mineralogy is always of the M-type.

The A-component (Fig. 5)

Another coherent high blocking temperature (500° up to 560°) and coercivity (30 up to 80 mT) component, with a shallow intermediate upwards inclination and a SSW declination, is separated in nearly all the samples from site DB (e.g. 86DB011A, 86DB016C, 86DB018C) and from site DN. In site DA, this component, apparently well isolated by A.F. demagnetization, does not clearly appear with thermal treatment (e.g. 86DA001A).

Host rock of dike-site DB was sampled for a baked contact test (site CDB). The magnetization component of the unbaked host rock is $D = 232$, $I = 59$, $k = 35$, $\alpha_{95} = 7$ (Ruffet et al., in prep.). In contrast, near the dike contact (e.g. 88CDB04A) the characteristic magnetization direction of the host rock is in close agreement with the results

TABLE 1

Palaeomagnetic results from Trégor dolerite dikes

	A-component						B-component						C-component					
	TH.			A.F.			TH.			A.F.			TH.			A.F.		
Site	D/I	n/n0	k	D/I	n/n0	k	D/I	n/n0	k	D/I	n/n0	k	D/I	n/n0	k	D/I	n/n0	k
DA				*214/-14*	*5/9*	*327*	217/3	8/8	236	218/2	7/9	64						
DB	194/-34	7/8	31	*216/-14*	*8/9*	*49*	219/5	8/8	90	209/3	6/9	57						
DC																172/60	8/9	53
DD							*195/28*	*7/8*	*27*	*193/31*	*7/10*	*28*				174/55	8/10	40
DE							217/13	3/8	46	209/14	3/5	48						
DF							211/3	8/8	49	213/9	8/9	53	*213/27*	*3/8*	*31*			
DG																		
DH							211/10	11/11	29									
DI							*180/-7*	*9/9*	*114*									
DJ							205/15	7/7										
DK							222/8	3/11	350									
DL							229/15	6/7	24									
DM																		
DN	191/-28	6/6	21				214/7	5/6	75									
DO							200/22	8/8	18	212/30	3/3	19						
DP																		
DQ							219/-1	6/6	484									
DR													186/52	4/5	113			
CDO							207/12	12/13	81									
CDB	189/-37	13/13	30				211/12	12/13	38									
CDR													169/35	4/4	332			
CDG													178/40	7/7	23			

	D/I	α_{95}	n/N	k	D/I	α_{95}	n/N	k	D/I	α_{95}	n/N	k
All sites	191/-33	8	26/3	237	214/10	4	124/13	65	174/50	9	31/5	59
Palaeo-pole	58°S,336°E dp=5°,dm=9°				29°S,318°E dp=8°,dm=2°				11°S,2°E dp=16°,dm=5°			
Palaeo-latitude	17.9°N				5.2°S				30.4°S			

Summary of the site-mean and group-mean palaeomagnetic directions. D = declination; I = inclination; TH. = thermal treatment; A.F. = alternating field treatment; n = number of statistical entries of sample directions used in the statistical analysis; n_0 = number of samples treated; N = number of statistical entries of sites in the statistical analysis; k = precision parameter (McElhinny, 1973); α_{95} = semi-angle of the cone of confidence of 95% probability (McElhinny, 1973).

obtained by thermal treatment, from the site DB doleritic dike. So, the baked contact test is positive.

However, the mean directions obtained by A.F. treatment differ from those obtained by thermal demagnetization. This feature results from an imperfect separation between the high-coercivity component and a lower-coercivity magnetization component (B-component, see below). The mean direction of the A-component, computed from

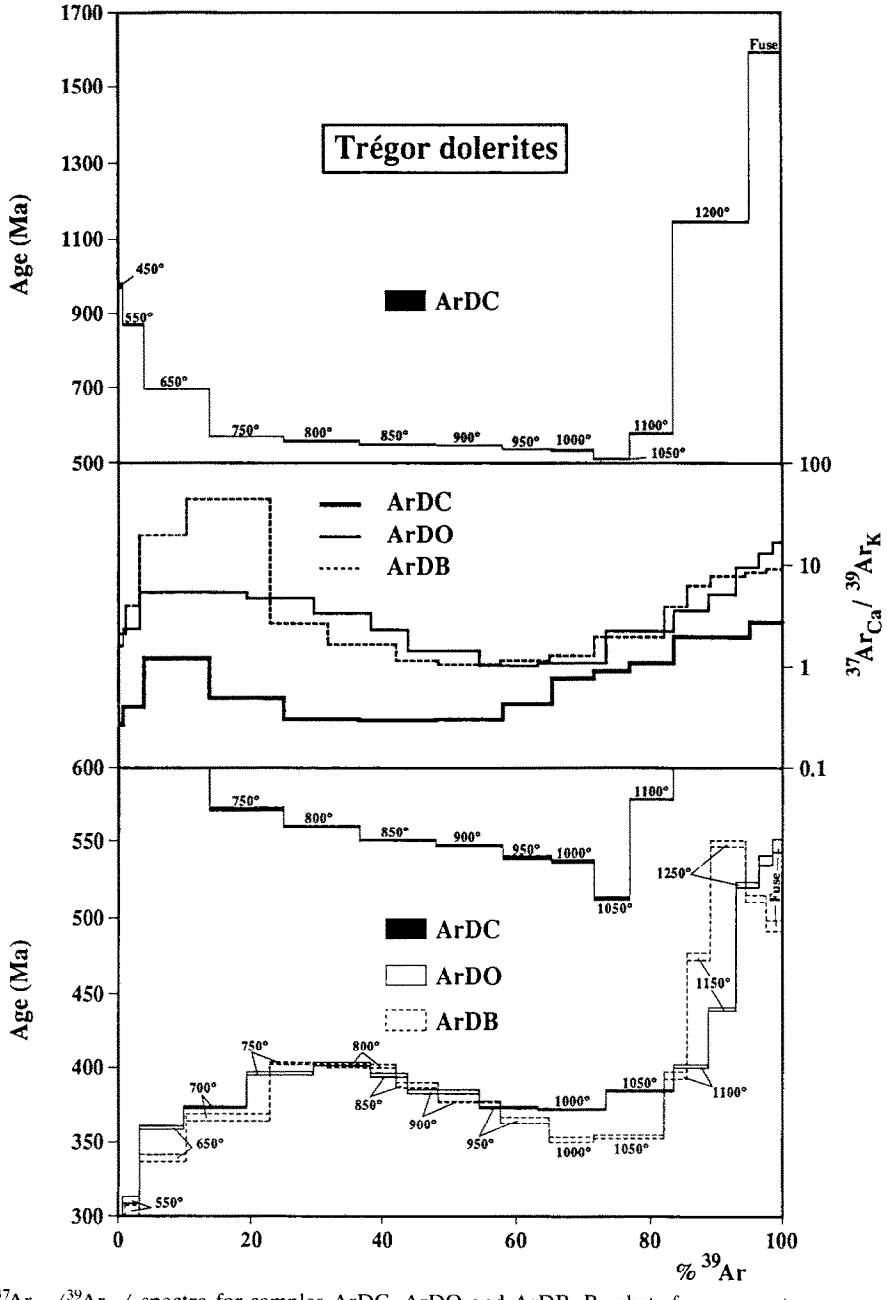


Fig. 10. Age and $^{37}Ar_{Ca}/^{39}Ar_K$/ spectra for samples ArDC, ArDO and ArDB. Brackets for apparent ages represent 1-σ errors. Temperatures are given in °C.

TABLE 2

^{40}Ar–^{39}Ar analytical data

T (°C)	Atmospheric (%)	$^{39}Ar_K$ (%)	$^{37}Ar_{Ca}/^{39}Ar_K$	$^{40}Ar/^{39}Ar_K$	Age (Ma)
ArDC plagioclase					
450	69.22	0.87	0.2710	49.45	974.5 ± 9.2
550	29.91	2.88	0.3960	42.69	868.4 ± 2.2
650	17.08	9.72	1.2045	32.55	696.9 ± 1.0
750	10.04	11.49	0.4817	25.79	572.6 ± 1.2
800	3.98	11.53	0.3010	25.18	560.9 ± 1.0
850	3.43	11.32	0.2953	24.68	551.3 ± 0.9
900	4.53	10.32	0.3046	24.47	547.3 ± 0.8
950	6.52	7.36	0.4260	24.06	539.4 ± 1.0
1000	9.18	6.07	0.7869	23.95	537.3 ± 1.1
1050	11.25	5.40	0.9219	22.67	512.1 ± 1.5
1100	12.47	6.63	1.0966	26.13	579.0 ± 1.1
1200	13.18	11.43	1.9613	61.23	1145.3 ± 2.3
Fuse	16.48	4.97	2.7500	97.72	1590.9 ± 2.5
					Aver. 699.0 ± 0.4
ArDC plagioclase laser fusion					
Fuse	6.40	–	0.7754	35.36	748 ± 21
ArDO plagioclase					
450	92.86	0.59	1.5964	8.89	219 ± 16
550	39.03	2.63	2.3482	13.00	311.2 ± 2.5
650	22.82	6.74	5.4883	15.25	360.1 ± 1.3
700	15.74	9.34	5.4288	15.90	373.9 ± 1.1
750	6.10	9.94	4.6711	16.97	396.5 ± 0.9
800	6.60	8.70	3.3319	17.24	402.2 ± 0.9
850	8.12	5.61	2.3218	16.92	395.4 ± 1.2
900	6.83	10.91	1.4227	16.39	384.4 ± 0.7
950	7.44	8.87	1.0405	15.89	373.7 ± 0.7
1000	6.87	10.20	1.0829	15.85	372.9 ± 0.8
1050	8.47	10.09	2.2387	16.42	385.0 ± 0.7
1100	12.70	5.12	3.4844	17.21	401.5 ± 1.1
1150	14.85	4.13	5.0883	19.05	439.5 ± 1.5
1250	17.16	3.55	9.4949	23.19	522.5 ± 1.9
1350	25.45	2.16	13.3760	23.99	538.0 ± 2.6
Fuse	35.25	1.42	16.9320	24.49	547.6 ± 3.7
					Aver. 393.9 ± 0.3
ArDO plagioclase laser fusions					
Fuse	18.86	–	2.3828	16.49	387 ± 64
Fuse	0.75	–	3.3423	21.77	494 ± 43
Fuse	8.34	–	4.0485	18.84	435 ± 57
ArDB plagioclase					
450	91.80	0.88	2.1076	8.73	215 ± 12
550	59.93	2.15	3.9702	12.65	303.5 ± 3.8
650	41.14	7.20	20.1710	14.31	339.9 ± 2.1
700	30.39	12.56	44.1510	15.60	367.4 ± 2.9
750	11.35	8.62	2.6107	17.28	402.9 ± 1.1
800	14.45	10.41	1.6305	17.20	401.3 ± 1.1
850	17.01	6.39	1.1563	16.59	388.4 ± 1.7
900	13.42	9.65	1.0525	16.08	377.6 ± 1.1
950	16.17	7.34	1.1635	15.51	365.7 ± 1.4

TABLE 2 (continued)

T (°C)	Atmospheric (%)	$^{39}Ar_K$ (%)	$^{37}Ar_{Ca}/^{39}Ar_K$	$^{40}Ar/^{39}Ar_K$	Age (Ma)
ArDB plagioclase (continued)					
1000	18.52	6.53	1.2911	14.89	352.2 ± 1.6
1050	19.29	10.46	1.9358	15.00	354.6 ± 1.1
1110	30.64	3.64	3.9157	16.92	395.5 ± 2.3
1150	35.18	3.42	6.2721	20.79	474.8 ± 2.4
1250	29.93	5.16	7.7228	24.56	549.0 ± 2.0
1350	33.29	3.26	8.4645	22.72	513.3 ± 2.3
Fuse	44.33	2.34	9.1001	21.80	495.1 ± 3.9
					Aver. 391.0 ± 0.5

$^{40}Ar/^{39}Ar$ analytical data for incremental heating experiments on samples ArDC, ArDO and ArDB, and laser single-step fusion experiments on pieces of plagioclase from samples ArDC and ArDO. ^{37}Ar is corrected for decay. Uncertainties are quoted at the level of one standard error. The ratios used to correct interfering isotopes are listed in Féraud et al. (1982). Isotopic constants used are those suggested by Steiger and Jäger (1977).

the thermally separated components is $D = 191.4$, $I = -32.9$, $k = 237.2$, $\alpha_{95} = 8.0$. The corresponding magnetic mineralogy is always of the P-type.

The B-component

The B-component (Figs. 4–6), with a shallow inclination and a SSW declination, is the more common magnetization component of the Trégor doleritic dikes. Regardless of the low-temperature magnetization, the B-component appears: (1) alone in site DE, DH (e.g. 86DH068B, Fig. 6), DJ, DK, DL and DQ (e.g. 88DQ18C, Fig. 6) with unblocking temperatures up to 570 °C or to high coercivities; (2) with the A- or C-component at intermediate coercivities or with a temperature spectrum bracketed between 300 ° and 500 °C in sites DB (e.g. 86DB011A, 86DB016C and 86DB018C, Fig. 5), DD (e.g. 86DD030B and C, Fig. 4), DF (e.g. 86DF046A, Fig. 6), DN and CDB (e.g. 88CDB04A, Fig. 5); or (3) alone with unblocking temperatures up to 550 °C but with magnetizations that do not decay towards the origin of the Zijdervelds diagrams in sites DA (e.g. 86DA001A, Fig. 5) and DO (e.g. 88DO03A, Fig. 6).

Host rock of dike-site DO was sampled for a baked contact test (site CDO). The magnetization component of the unbaked host rock is similar to the one previously quoted, whereas at contact (e.g. 88CDO03A) it is similar to the direction obtained from the site DO doleritic dike, with unblocking temperature up to 570–580 °C. Nevertheless, as observed for some samples from site DO as well, the demagnetization trajectories generally do not go to the origin in the Zijderveld diagrams. The residual extrapolated magnetization component defined in this way, $D = 217.8$, $I = 39.6$, $k = 23.6$, $\alpha_{95} = 14.1$, differs from the characteristic component of (1) the site DO doleritic dike, and (2) the unbaked host rock. So, the baked contact test is negative.

The apparent secondary magnetization of site DD, with a higher inclination than those generally obtained for the B-component in other sites, probably results from an imperfect separation and was not used in computation of the mean direction. The mean direction of the B-component, computed from (1) the eleven doleritic dike-site means (DA, DB, DE, DF, DH, DJ, DK, DL, DN, DO, DQ), (2) the baked contact site mean (CDO), and (3) from the mean of the lower blocking temperature or coercivity magnetization of the baked contact site CDB, is $D = 214.1$, $I = 9.8$, $k = 68.8$, $\alpha_{95} = 5.0$. Both P- and M-type magnetic mineralogies characterize the B-component.

Geochronological study

Sampling

Three hand specimens of dolerites were sampled for geochronology at sites DB (Sample ArDB), DC (Sample ArDC) and DO (Sample

ArDO) where the characteristic magnetization components have been identified (Fig. 1).

Results

The results are summarized in Table 2. Errors are quoted at the 1σ level and do not include the errors on the $^{40}Ar^*/^{39}Ar_K$ ratio and age of the monitor.

Sample ArDO

The age spectrum (Fig. 10) yields an integrated age of 393.9 ± 0.3 Ma. After a slow increase of the ages from 219 ± 16 Ma (step 450 °C) up to 402.2 ± 0.9 Ma (step 800 °C) for the first 38% of $^{39}Ar_K$ released, the age spectrum shows a regular decrease of the apparent ages up to 372.9 ± 0.8 Ma followed by a rapid increase up to 547.6 ± 3.7 Ma. Between 800 °C and fusion, the $^{37}Ar_{Ca}/^{39}Ar_K$ and age spectra are directly correlated. The highest apparent ages of the high-temperature steps (547.6 ± 3.7 Ma at fusion step) correspond to the highest values of $^{37}Ar_{Ca}/^{39}Ar_K$ ratios (up to 16.9) (Ca/K = 30.2; from the relation $^{37}Ar_{Ca}/^{39}Ar_K = 0.56 \times Ca/K$; Féraud et al., 1982), whereas the lowest ages (373.7 ± 0.7 Ma and 372.9 ± 0.8 Ma at 950 and 1000 °C, respectively) are characterized by the lowest $^{37}Ar_{Ca}/^{39}Ar_K$ measured ratios (1.0 and 1.1, respectively, Ca/K ~ 1.8). For steps 650 and 700 °C, there was probably degassing of calcite.

Sample ArDB

The age spectrum (Fig. 10) yields an integrated age of 391.0 ± 0.5 Ma. It has a shape very similar to the one displayed by sample ArDO. There is an important degassing of calcite at steps 650 and 700 °C, characterized by high $^{37}Ar_{Ca}/^{39}Ar_K$ ratios (up to 44.2). The maximum ages reached at 750 and 800 °C (402.9 ± 1.1 Ma and 401.3 ± 1.1 Ma respectively) are undistinguishable from the age of 402.2 ± 0.9 Ma obtained from sample ArDO. The minimum ages of the "saddle" at intermediate-temperature steps (352.2 ± 1.6 Ma and 354.6 ± 1.1 Ma at steps 1000 and 1050 °C respectively) are lower than those obtained on sample ArDO. Unlike experiment ArDO, the minimum ages of experiment ArDB do not correspond exactly to the minimum values of the $^{37}Ar_{Ca}/^{39}Ar_K$ spectrum. The high-temperature ages reach 549.0 ± 2.0 Ma (step 1250 °C) and then decrease to 495.1 ± 3.9 Ma. These high ages also correspond to the highest $^{37}Ar_{Ca}/^{39}Ar_K$ measured ratios.

Sample ArDC

The age spectrum (Fig. 10) displays a saddle-shape characteristic of excess ^{40}Ar (Lanphere and Dalrymple, 1976; Harrison and McDougall, 1981). The ages decrease rapidly from 974.5 ± 9.2 Ma (step 450 °C) to 572.6 ± 1.2 Ma (step 750 °C), then more slowly to reach a minimum value of 513.1 ± 1.5 Ma (1050 °C). In the high-temperature steps, the ages climb very rapidly up to 1590.9 ± 2.5 Ma. For the saddle-shaped age spectra, the minimum age is generally interpreted as a maximum estimate of the true age (Lanphere and Dalrymple, 1976). The $^{37}Ar_{Ca}/^{39}Ar_K$ spectrum displays also a saddle-shape with a minimum value of 0.3 that does not correspond, as for sample ArDB, to the minimum age of the age spectrum.

Laser single step fusion

Total laser fusions were performed on pieces of plagioclase. Three pieces of plagioclase from sample ArDO have displayed ages of 387 ± 64 Ma, 435 ± 57 Ma and 494 ± 43 Ma with corresponding $^{37}Ar_{Ca}/^{39}Ar_K$ ratios of 2.4, 4.1 and 3.3, respectively. One piece of plagioclase from sample ArDB has provided an age of 748 ± 21 Ma with a respective $^{37}Ar_{Ca}/^{39}Ar_K$ ratio of 0.78.

Correlation calculations

The correlation calculation ($^{36}Ar/^{40}Ar$ versus $^{39}Ar/^{40}Ar$; Turner, 1971; Roddick et al., 1980; Hanes et al., 1985), an alternative form to the classical isochron analysis ($^{40}Ar/^{36}Ar$ versus $^{39}Ar/^{36}Ar$; Merrihue and Turner, 1966) has been performed from the results of experiments ArDC and ArDB. The points plotted in the correlation diagrams (Fig. 11) show too much scatter to define correct linear arrays; therefore it is impossible to correctly distinguish the various components of trapped argon.

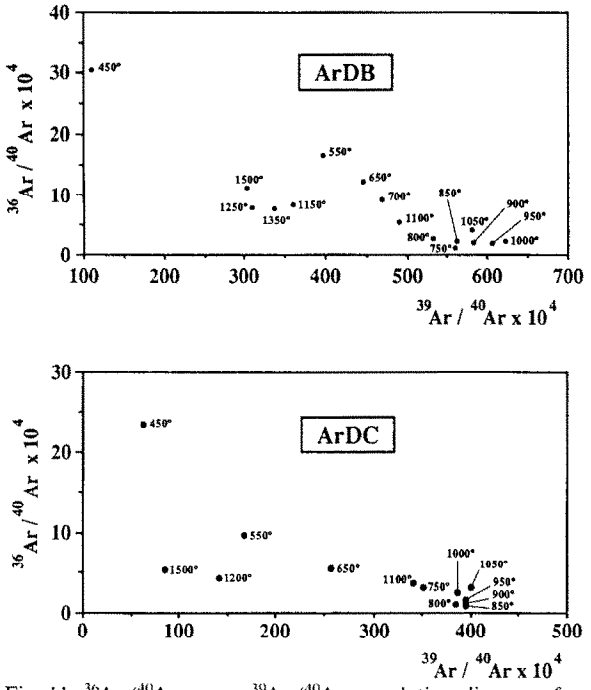


Fig. 11. $^{36}Ar/^{40}Ar$ versus $^{39}Ar/^{40}Ar$ correlation diagrams for samples ArDC and ArDB.

Discussion of the results

At site DD, the unblocking temperature of the C-component is higher than that of the B-component. Similarly, in sites DB, DN and CDB, the unblocking temperature of the A-component is higher than that of the B-component. All these facts, with the negative baked contact test at site CDO, suggest that the B-component is a remagnetization of younger age than the A or C magnetization components (Fig. 12).

The baked contact test performed for components A (site CDB) and C (sites CDR and CDG) confirmed that these two components are primary magnetizations, assumed furthermore from their behaviour during thermal and A.F. demagnetizations. Unfortunately, the total absence of relationships between these two components of magnetization preclude an attempt of relative dating.

The B-component of remagnetization is the

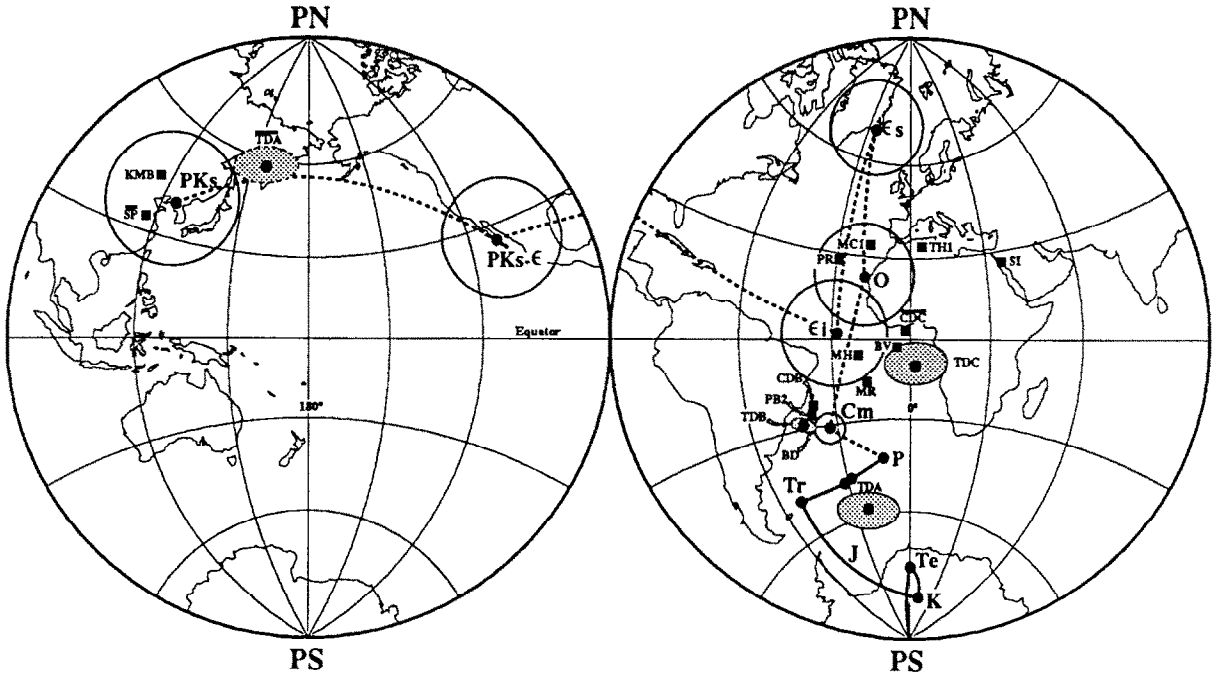


Fig. 12. Apparent polar wander path (stereographic projection) for the Armorican Massif from Upper Precambrian till Recent (from Perroud, 1985). *KMB* = Tréguier kératophyres (Perigo et al., 1983); *SP* = Paimpol spillites (Hagstrum et al., 1980); *MH* = Loguivy microgranite (Perigo et al., 1983); *MR* = Lézardrieux/St Germain rhyolites (Hagstrum et al., 1980); *CDC* = Crozon dolerites (Perroud et al., 1983); *BV* = Builth Volcanic Series (Briden and Mullan, 1984); *SI* = Shelve Inlier volcanic rocks (McCabe and Channell, 1990); *MC1* = "Moulin de Châteaupanne" (Perroud et al., 1986b); *ES* = Erquy spilites (Duff, 1979); *PR* = Pont Réan formation (Cogné, 1988); *TH1* = Thouars Massif (Perroud and Van der Voo, 1985); *BD* = St Malo dolerites (Perroud et al., 1986a); *CDB* = Crozon dolerites (Perroud et al., 1983); *PB2* = Paimpol-Brehec volcano sedimentary series (Jones et al., 1979).

most widely represented one (124 directions from thirteen sites). It is well defined with high within-site precision parameters (k). The mean direction yields a palaeopole (labelled TDB) at 29° S, 318° E. It is close to the mean of the Armorican Massif palaeopoles for the interval 280–330 Ma compiled by Perroud (1985) and more specially with the Lower Carboniferous palaeopole from the St Malo doleritic dike swarm (Perroud et al., 1986a) and with the palaeopole corresponding to the Hercynian remagnetization isolated (1) in the Cambro–Ordovician volcano–sedimentary series of Paimpol-Bréhec (Trégor area; Jones et al., 1979), and (2) in the Ordovician Crozon dolerites (Perroud et al., 1983). This agreement leads us to presume that the palaeopole TDB is likely to be of Carboniferous age.

Geochronological sample ArDO comes from the dike-site DO, where the B-component was observed. The very disturbed shape of the age spectrum does not allow to constrain the age of the B-component (assumed, from palaeomagnetic results, to be Carboniferous), or the age of the dike. We propose that the high ages obtained at high-temperature steps reflect the presence of a component of argon in excess. Unfortunately, the single step laser fusions of pieces of plagioclase from sample ArDO do not help to test this hypothesis.

The C-component (31 directions from five sites) yields a palaeopole (labelled TDC) at 11° S, 2° E and a palaeolatitude of 30° S. The comparison of this palaeolatitude with the curve of variations of palaeolatitudes for northern Spain from Ordovician to Carboniferous times (Perroud et al., 1991) leads us to propose that the C-component is Upper Ordovician in age (Fig. 13). This interpretation must be viewed tentatively because we have not taken account of errors in the mid-Palaeozoic section of our reference curve. If our interpretation is correct then the palaeopole TDC appears to be quite different from: (1) the Lower Ordovician palaeopoles from (a) the Moulin de Châteaupanne (Arenig; Perroud et al., 1986b), (b) the red series of the Pont-Réan formation (Cogné, 1987); and (2) the Late Ordovician palaeopole from (a) the Thouars Massif (444 ± 9 Ma,

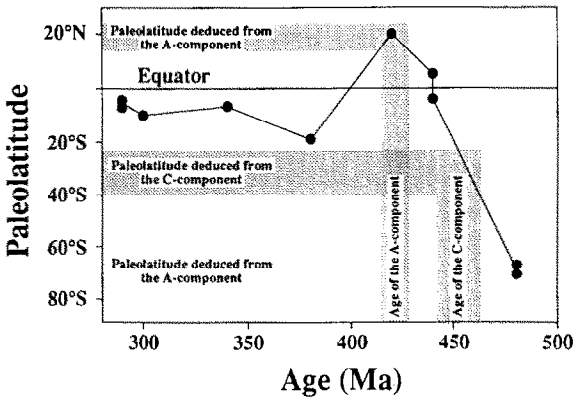


Fig. 13. Comparison of the palaeolatitudes deduced from the A- and C-components with the curve of variations of the palaeolatitudes for northern Spain from the Ordovician to Carboniferous (from Perroud et al., 1990).

Bernard-Griffiths and Le Métour, 1979) (Perroud and Van der Voo, 1985), and (b) the Shelve Inlier (Wales; McCabe and Channell, 1990). On the other hand, the palaeopole TDC is in close agreement with the palaeopole obtained from the upper Ordovician Crozon dolerites (Perroud et al., 1983) or with the one obtained by Briden and Mullan (1984) from the Builth volcanic series of Llanvirnian age, mid-Wales, and more generally with Ordovician palaeopoles of the British Isles from the south of the Iapetus suture as compiled by Briden et al. (1984).

The palaeopole TDC is also in rather good agreement with the Eocambrian palaeopoles of the Armorican Massif such as the Loguivy Microgranite (Perigo et al., 1983) or the Lézardrieux/St Germain Rhyolites (Hagstrum et al., 1980) palaeopoles.

The geochronological sample ArDC comes from the site-dike DC, where we observed the C-component. The saddle-shape of the age spectrum indicates the presence of a trapped component of ^{40}Ar in excess. This excess argon is confirmed by the anomalously high ages of 748 ± 21 Ma, older than the country rock (615 ± 15 Ma; Graviou, 1984), obtained by laser single step fusion of a piece of plagioclase. Nevertheless, the correlation calculation does not confirm this excess of ^{40}Ar, on account of too much scatter of points in the correlation diagram.

The plagioclase is frequently saussuritized probably in the presence of K-rich alteration

products such as sericite. Assuming that the destabilization of the plagioclase is contemporaneous with intrusion of the dike (Auvray, 1979), the minimum age of the saddle (513.1 ± 1.5 Ma) is a maximum estimate of the true age of the dike (Lanphere and Dalrymple, 1976). This assumption is reinforced by the fact that the minimum age of the saddle does not correspond to the minimum value of the $^{37}Ar_{Ca}/^{39}Ar_K$ spectrum. The upper limit fixed by this minimum age implies that the C-component probably does not have an Eocambrian age (540–550 Ma) but rather an Upper Ordovician age.

Unlike the B- and C-components which are not geographically constrained, the A-component (26 directions from three sites) appears only in the doleritic dikes of the "Sillon de Talberg" area. It yields a palaeopole (labelled TDA) at 58° S, 336° E that appears quite different from all previously reported Palaeozoic palaeopoles for Brittany. On the other hand, it appears rather in better agreement with (1) the Tréguier keratophyres (Perigo et al., 1983) and the Paimpol spilites (Hagstrum, 1980), both dated at 642 ± 46 Ma (Vidal, 1980), and (2) the Permo–Triassic palaeopoles for Europe. These agreements could lead us to presume that the palaeopole TDA is either of Precambrian (Cadomian) or Permo–Triassic age. These two possibilities seem unlikely because in case (1) the Trégor doleritic dikes, with continental affinities (Auvray, 1979), would be of the same age as the host rock, the calc-alkaline North Trégor batholith (615 ± 15 Ma; Graviou, 1984; Graviou et al., 1989) related to the North Armorican Cadomian subduction zone (Graviou et al., 1989), and in case (2) the A-component, older than the B-component of Carboniferous age, cannot be of Permo–Triassic age.

The geochronological sample ArDB comes from the dike-site DB, where we obtained a positive baked contact test. The age spectrum of sample ArDB, similar to the one of sample ArDO, is interpreted in the same way, assuming the presence of a trapped component of ^{40}Ar in excess. Nevertheless, as for sample ArDC, the correlation calculation does not confirm this excess of ^{40}Ar, because there is too much scatter in the points in the correlation diagram.

As for the other geochronological samples, the plagioclase from sample ArDB is frequently saussuritized with the appearance of alteration K-rich phases such as sericite. These alteration phases could be related to the Hercynian partial remagnetization observed on all the palaeomagnetic samples from site DB. As for sample ArDC, the minimum ages of the saddle-shaped part of the age spectrum (steps 1000° and 1050° C) do not correspond to the minimum measured values of the $^{37}Ar_{Ca}/^{39}Ar_K$ spectrum, reinforcing the assumption of ^{40}Ar in excess, at the origin of the high ages obtained in the high-temperature steps. The disturbed shape of the age spectrum does not allow to further constrain the age of the A-component.

In the same way as for the C-component, the comparison of the palaeolatitude deduced from the A-component with the curve of variations of the palaeolatitudes for northern Spain from Ordovician to Carboniferous times (Perroud et al., 1991) suggests an Upper Silurian age (Fig. 13). This assumption is based on the palaeolatitude of 20° N obtained by Perroud et al. (1991) for the Upper Silurian (420 Ma) volcanism of the Almaden mining district (southern Spain). If we suppose Gondwanian affinities, assumed from palaeontological criteria (Van der Voo, 1988), a palaeolatitude of 17.9° N (deduced from the A-component) for the northern part of the Armorican Massif in Upper Silurian supports the model of an apparent polar wander path of Gondwana between Ordovician and Devonian proposed by Van der Voo (1988) and Bachtadse and Briden (1990). This APWP rests on the African palaeopole of the Air intrusives (Niger) (435 Ma; Hargraves et al., 1987) and the Australian palaeopole for the Snowy River volcanics (Lower Devonian; Schmidt et al., 1986) when rotated according to the Gondwana fit of Smith and Hallam (1970). Effectively this path makes a loop between the Ordovician pole (located in the northwestern part of Africa) and the Devonian pole (located in Central Africa), with its apex, at Upper Silurian, located to the southwest of South Africa. This implies an equatorial position for the North African part of Gondwana and then an incursion, for the first time in the Palaeozoic, of

the Armorican Massif into the Northern Hemisphere.

Geodynamical implications and conclusions

The geochronological study has not been successful but nevertheless suggests that the intrusive phases occur during Palaeozoic times.

The palaeomagnetic analyses of the Trégor dolerite dikes have revealed two apparently primary and one secondary magnetization component. The direction of the secondary component clearly links its acquisition to the well-known late Palaeozoic remagnetization event. Its perfect concordance with all Carboniferous results from the Armorican Massif shows that no local rotation has occurred in the Trégor area since its acquisition. The presence of two primary components seems more difficult to interpret. A first explanation can be differential structural tilting of the area between the magnetization acquisitions. This would require at least 90° of tilt, which is not supported by geological evidence (the Variscan orogeny is restricted in this area to minor faulting without noticeable local rotations) or by the palaeomagnetic results from the host rock (Ruffet et al., in prep.). Other structural interpretations incorporating vertical axis rotations and polarity changes are not plausible for the same reasons. It seems far more reasonable to assume that the magnetizations are not synchronous, either because there were two intrusion phases or because the cooling through Curie temperatures was not synchronous over the whole area. Given the high apparent plate velocity suggested by Gondwanian palaeomagnetic data (Van der Voo, 1988; Bachtadse and Briden, 1990) or Spanish results (Perroud et al., 1991), a small time lag in remanence acquisition would lead to a significative change of the field direction. So, it does not seem unrealistic to assume Upper Ordovician and Upper Silurian ages for the C and A magnetization components, respectively.

Using these two new palaeopoles, we have modified the Armorican Massif APWP proposed by Perroud (1985) for the Ordovician–Carboniferous time interval (Fig. 14). Furthermore, we have calculated an alternate APWP, corrected for

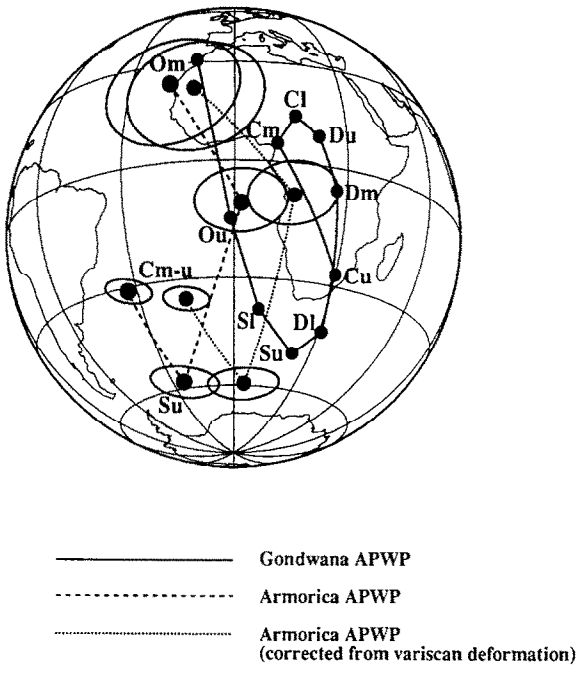


Fig. 14. Comparison of the APWP of Armorica from Ordovician to Carboniferous with that of Gondwana. See text for explanation.

the Variscan 15° counterclockwise rotation revealed by comparison with the stable Europe Middle–Upper Carboniferous reference pole (Perroud, 1986).

The comparison of these new APWPs with the Gondwana one, as proposed by Bachtadse and Briden (1990), shows a relatively good fit of the Ordovician–Upper Silurian segments of the paths, followed by a clear divergence of the paths in Upper Palaeozoic. The timing of the divergence cannot be precisely established since no Devonian result is available for the Armorican Massif.

Perroud et al. (1984) have proposed that the Armorica plate was linked to the northern margin of Gondwana during Ordovician time, in very high southern latitudes. The similarity of the palaeolatitudes of North Africa and Armorica in Upper Silurian suggests that this conclusion can be extended to the whole Ordovician–Silurian time interval, even if there exist minor discrepancies between the polar wander paths. Thus, it seems that Armorica was separated from Gondwana sometime during the Devonian or Carboniferous, since the Upper Carboniferous poles are very different.

These palaeogeographical implications rely strongly upon the age of the palaeopoles, which were determined indirectly by comparison with previous palaeomagnetic results. This situation is not satisfying, and limits their reliability. To overcome this problem, new geochronological investigations have been started, with the hope that better constraints on the dike-emplacement ages will be obtained.

Acknowledgments

We are grateful to Prof. J.J. Chauvel for his help in the optical examination of polished thin sections. We also thank Dr. L. Harris and two anonymous reviewers for their critical reading and correction of the English. This work has benefited of financial support from CNRS-INSU through its national program DBT (contribution No. 222).

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